

2 Description

The STM32C011x4/x6 mainstream microcontrollers are based on high-performance Arm® Cortex®-M0+ 32-bit RISC core operating at up to 48 MHz frequency. Offering a high level of integration, they are suitable for a wide range of applications in consumer, industrial and appliance domains and ready for the Internet of Things (IoT) solutions.

The devices incorporate a memory protection unit (MPU), high-speed embedded memories (6 Kbytes of SRAM and up to 32 Kbytes of flash program memory with read and write protection, proprietary code protection, and securable area), DMA, an extensive range of system functions, enhanced I/Os, and peripherals. The devices offer standard communication interfaces (one I²Cs, one SPI / one I²S, and two USARTs), one 12-bit ADC (2.5 MSps) with up to 15 channels, a low-power RTC, an advanced control PWM timer, four general-purpose 16-bit timers, two watchdog timers, and a SysTick timer.

The devices operate within ambient temperatures from -40 to 125°C and with supply voltages from 2.0 V to 3.6 V. Optimized dynamic consumption combined with power-saving modes allows the design of low-power applications.

The devices are housed in packages with 8 to 20 pins.

Table 2. STM32C011x4/x6 family device features and peripheral counts

Peripheral		STM32C011_				
		_J4	_J6	_D6	_F4	_F6
Flash memory (Kbyte)		16	32	32	16	32
SRAM (Kbyte)		6 with parity				
Timers	Advanced control	1 (16-bit)				
	General-purpose	4 (16-bit)				
	SysTick	1				
	Watchdog	2				
Comm. interfaces	SPI [I2S] ⁽¹⁾	1 [1] + 2 extra through USARTs				
	I2C	1				
	USART	2				
RTC / RNG / AES / VREFBUF		Yes / No / No / No				
GPIOs (all 5V-tolerant)		6		10	18	
DMA channels		3				
Wakeup pins		2		4		
12-bit ADC channels (external + internal)		5 + 2		7 + 2	13 + 2	
Max. CPU frequency		48 MHz				
Operating voltage		2.0 to 3.6 V				
Operating temperature ⁽²⁾		Ambient: -40 to 85 °C / -40 to 105 °C / -40 to 125 °C				
Package		SO8N		WLCSP12	UFQFPN20	
Bootloader		USART1, I2C1				

1. The number in brackets denotes the count of SPI interfaces configurable as I2S interface.

2. Depends on order code. Refer to [Section 7: Ordering information](#) for details.